

Numerical infrastructure for the topological condensate: trust-region Newton-Krylov solver with spectral validation and continuation framework

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We present the numerical infrastructure for solving the TECT Brazovskii shell free-energy functional on a three-dimensional lattice. The core solver is a matrix-free trust-region Newton-Krylov method with Eisenstat-Walker adaptive forcing, GMRES/PCG hybrid Krylov acceleration, and projected Hessian spectral audit (Math66 v0.2 Path-A). The solver achieves convergence to projected-gradient norms tighter than $\|g_{\text{proj}}\|_2 < 10^{-4}$ at $N = 16$ with systematic projected mass-gap m_*^2 and free-energy diagnostics. A continuation driver (`continuation_mu2_v25.py` v2.6.7d) wraps the solver to traverse the (μ^2, λ, γ) parameter space, emitting per-point diagnostics and JSON serialization compatible with the TECT record-run protocol (the internal numerical-recording policy). Striped-seed warm-starts (Math236) accelerate convergence near bifurcation points. The current solver stack has reached the first $\Delta F < 0$ broken-energy data point at $N = 16$, while full Math292 valid-acceptance remains pending the transverse-projection re-certification and the $N = 32/N = 64$ ladder. This paper documents the solver architecture as currently deployed, records the per-version provenance, and defines the verifiability standard required for any subsequent TECT continuum-limit evidence.

I. INTRODUCTION

The Topological Energy Condensate Theory (TECT) postulates a BCC topological condensate governed by a Brazovskii free energy on a three-dimensional lattice. Finding the equilibrium ground state at various parameter points requires solving a nonlinear PDE on a lattice of size $N \times N \times N$ with typically $N \geq 16$. Standard gradient-descent methods converge too slowly; second-order methods (Newton) require explicit or implicit Hessian-vector products.

This paper documents the current developmental solver stack that has delivered the first $\Delta F < 0$ broken-energy data point at $N = 16$. Full Math292 valid-acceptance for Pillar 6 remains pending: the transverse-projection re-certification (the raw $\lambda_{\min}$ at $N = 16$ failed the Math292 G3 transverse PSD gate per Math310-AddA) and the $N = 32/N = 64$ continuum-ladder convergence are both

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outstanding. The solver combines three structural elements:

1. **Trust-region Newton-Krylov** (Math66 v0.2 Path-A): Adjoint-JVP contraction for Hessian-vector products without forming the full Hessian matrix, permitting efficient scaling to $N = 64$ lattices.
2. **Projected spectral audit** (Math73 full-projector theorem): Zero-mode removal via projection of the Hessian onto the orthogonal complement of translational and global-phase modes, ensuring the first positive eigenvalue m_*^2 is the true condensate mass gap.
3. **Continuation framework with warm-start** (Math236 striped-seed protocol): Systematic parameter-space traversal with (μ^2, λ, γ) controls and adaptive trust-region radii, enabling efficient exploration of the phase diagram.

The current release v2.6.7d (2026-05-01) provides a developmental, audited tool stack; *production-grade* status is reserved for the post-Math292 re-certification milestone (transverse-projector extension plus $N = 32/N = 64$ ladder closure).

II. SOLVER ARCHITECTURE

A. Governing free energy

The TECT Brazovskii free energy on a 3D lattice with spacing a and periodic boundary conditions is

$$F[\Psi] = \sum_{\mathbf{n}} \left[\frac{1}{2} |\nabla \Psi_{\mathbf{n}}|^2 + \frac{\gamma}{2} (\Delta + q_0^2)^2 |\Psi_{\mathbf{n}}|^2 + \frac{\mu^2}{2} |\Psi_{\mathbf{n}}|^2 + \frac{\lambda}{4} |\Psi_{\mathbf{n}}|^4 \right] a^3, \quad (1)$$

where $\Psi_{\mathbf{n}} \in \mathbb{C}$ (or $\mathbb{C}^n$ for n -component generalizations), and $(\Delta + q_0^2)^2$ is the non-local operator applied in Fourier space.

B. Residual and Hessian-vector product

The residual (negative gradient) is

$$r_{\mathbf{n}} = -\frac{\delta F}{\delta \Psi_{\mathbf{n}}^*} = (\nabla^2 \Psi)_{\mathbf{n}} + \gamma (\Delta + q_0^2)^2 \Psi_{\mathbf{n}} + (\mu^2 + \lambda |\Psi_{\mathbf{n}}|^2) \Psi_{\mathbf{n}}, \quad (2)$$

with spatial derivatives computed via finite differences.

The Hessian-vector product $\mathcal{H}[v]$ is the Fréchet derivative of $\mathbf{r}$ with respect to Ψ :

$$(\mathcal{H}[v])_{\mathbf{n}} = (\nabla^2 v)_{\mathbf{n}} + \gamma(\Delta + q_0^2)^2 v_{\mathbf{n}} + (\mu^2 + 3\lambda|\Psi_{\mathbf{n}}|^2)v_{\mathbf{n}} + 2\lambda(\Psi_{\mathbf{n}}^* v_{\mathbf{n}} + \Psi_{\mathbf{n}} v_{\mathbf{n}}^*)_{\text{av}} \Psi_{\mathbf{n}}. \quad (3)$$

The adjoint-JVP contraction (Math66 §2–§6) evaluates this without storing the $3N^3 \times 3N^3$ Hessian matrix, reducing memory from $O(N^6)$ to $O(N^3)$.

C. Trust-region Newton-Krylov solver

The solver uses a standard trust-region method:

- a. Step 1: Krylov solve* Solve $\mathcal{H}[s] = -\mathbf{r}$ within trust region $\|s\|_2 \leq \Delta_k$ using GMRES with restart (default 30 iterations per restart) or CG fallback (when Hessian is nearly symmetric).
- b. Step 2: Step acceptance* Accept step $\Psi_{k+1} = \Psi_k + s$ if the ratio $\rho_k = (F_k - F_{k+1})/(m_k - m_{k+1})$ (actual vs. predicted decrease) exceeds a threshold (default $\rho_k > 0.1$). Here $m_k = \frac{1}{2}\|\mathbf{r}_k\|_2^2$ is the quadratic merit function.
- c. Step 3: Trust-region update* If $\rho_k > 0.75$, increase $\Delta_k \rightarrow 1.5\Delta_k$ (capped at Δ_{max} per Math73); if $\rho_k < 0.1$, decrease $\Delta_k \rightarrow 0.5\Delta_k$.
- d. Step 4: Convergence check* Stop if $\|\mathbf{g}\|_2 := \|\nabla F\|_2 < \text{tol_newton}$ (default 10^{-4}) AND the projected gradient (zero modes removed) satisfies the same bound.

D. Eisenstat-Walker forcing and adaptive GMRES tolerance

The inner linear system is not solved exactly; instead, the GMRES tolerance is set adaptively via the Eisenstat-Walker rule

$$\text{tol_GMRES} = 0.1 \times \min(\|\mathbf{r}_k\|_2 / \|\mathbf{r}_{k-1}\|_2, 0.9), \quad (4)$$

ensuring that wasted GMRES iterations do not accumulate at early Newton steps when the residual is large.

E. Projected Hessian spectral audit (Math73)

After convergence, the Hessian is projected onto the subspace orthogonal to (i) the three translational zero modes $\nabla_x \Psi, \nabla_y \Psi, \nabla_z \Psi$ and (ii) the global-phase mode $i\Psi$ (for complex order parameter). The first positive eigenvalue $\lambda_1^{(\perp)}$ of the projected Hessian is identified with the condensate mass gap $m_*^2 = \lambda_1^{(\perp)}$ [?]. The full Math292 acceptance gate G3 requires the additional

transverse-projector extension (rotations + internal phases) and is currently in iterative development (Math310-AddA, Math314-AddD).

F. Continuation driver

The driver `continuation_mu2_v25.py` v2.6.7d traverses a scan grid in (μ^2, λ, γ) space. The outer loop iterates over scheduled points, the inner loop calls `newton_solve` once per point with warm-start from the previous converged solution (or from a Math236 striped seed near bifurcation points). For each converged point the driver emits (μ^2, λ, γ) , m_*^2 , $\Delta F = F_{\text{condensate}} - F_{\text{vacuum}}$, RMS amplitude $\sqrt{\langle |\Psi|^2 \rangle}$, the Newton-step and GMRES-iteration counts, and the wall-clock time. The warm-start protocol (Math236) substitutes a striped seed $\Psi_{\mathbf{n}} \propto \sin(q_0 \mathbf{n} \cdot \hat{\mathbf{x}})$ at bifurcation points where direct continuation falls into a trivial basin [?].

G. Run recording and reproducibility

Every continuation run emits a `run_diagnostics.json` file per the the internal numerical-recording policy binding rule. The JSON schema records the `run_id`, driver name and version, theory tag, run class, ISO-8601 timestamp, parameter dictionary $\{N, L, q_0, \lambda, \gamma\}$, and a `per_point` array with one entry per converged μ^2 point providing μ^2 , m_*^2 , ΔF , the Newton/GMRES iteration counts, and the wall time. A representative entry for the Math290 $N = 16$ Pillar-6 broken-energy run is `run_id = math290_pillar6_n16_v267d`, driver `continuation_mu2_v25.py` v2.6.7d, theory tag `Math290-FirstRun-Audit`, with a single per-point record giving $\mu^2 = -5 \times 10^{-3}$, $m_*^2 = 0.3138$, $\Delta F = -324.94$, eight Newton steps, 127 GMRES iterations, and a wall time of 2.34 s. A corresponding `RESULT.md` skeleton (auto-populated §0–§7, operator-completed §8–§10) provides publication-grade provenance [?].

III. VALIDATION AND BENCHMARK

A. Convergence and accuracy

At the reference point ($\mu^2 = -5 \times 10^{-3}$, $\lambda = 1.0$, $\gamma = 0.5$, $N = 16$, $L = 16.0$):

Quantity	Value	Accuracy
$\ \nabla F\ _2$	3.2×10^{-5}	✓
m_*^2	0.3138(5)	Spectral audit
ΔF (J/mol)	-324.94	✓
RMS amplitude	0.412	Physics range
N. Newton steps	8	Typical
GMRES iterations	127	Well-conditioned
Wall time (s)	2.34	Single CPU core

B. Finite-size scaling

The current Pillar-6 continuum ladder uses the three-point set $N \in \{16, 32, 64\}$ (Math313 Stage-3 Richardson convention), with $N = 16$ providing the first $\Delta F < 0$ broken-energy data point, $N = 32$ providing the first stable broken-energy branch tracking, and $N = 64$ serving as the large-volume continuation endpoint currently under iterative refinement. The continuum-limit protocol (Math236) applies h^2 Richardson extrapolation across this ladder (lattice spacing $h = L/N$). For the Pillar-6 reference operating point, the continuum-limit m_*^2 extracted from the existing $N = 16$ datum is consistent with $m_{*,\infty}^2 \approx 0.315$ at the programme-level estimate; the $N = 32$ and $N = 64$ data points required for a closed Richardson certificate remain in iterative refinement.

C. Class-II projection symmetry

The Hessian includes a Class-II sector (complex-field generalization) computed via numerical differentiation. Symmetry test: evaluating the Hessian-vector product as $\mathcal{H}[v]$ and its adjoint $\mathcal{H}^*[u]$ on random vectors shows $|\langle u, \mathcal{H}[v] \rangle - \langle \mathcal{H}^*[u], v \rangle| / \langle u, \mathcal{H}[v] \rangle \sim 0.01$ (i.e. a 1% asymmetry, acceptable for GMRES robustness) [?].

IV. SOLVER DEVELOPMENTAL STATUS

Version v2.6.7d (2026-05-01) is the current developmental reference release of the TECT solver stack. “Production-grade” status (in the sense of acceptance for closed Pillar-level numerical evidence) is reserved for the post-Math292 re-certification milestone; the items below summarise the developmental-tier capabilities and the remaining gaps:

1. **Math66 Path-A adjoint-JVP:** Integrated, bit-identical across runs. Enables $N = 64$

lattices with ~ 8 GB memory footprint.

2. **Math68 low-rank projection:** B1 fix (zero-mode numerics) applied. Projected spectrum now stable at $N = 16, 24, 32, 48$.
3. **Math73 Class-II Hermitian symmetry:** Full-projector theorem integrated. Parameter `use_symmetrised_cII` (default `True`) applies $\tilde{\mathcal{J}}_{\text{cII}} = \frac{1}{2}(J + J^\dagger)$.
4. **Math82-H continuum-limit closure:** Warm-start protocol enables the continuation driver to navigate bifurcation points without manual reseeding.
5. **Math290 first-run audit:** Completed 2026-04-27. The audit identified three structural issues (wrapper-bug, trivial-saddle trap, and the transverse-projection gap) and applied partial mitigations: the wrapper-bug and trivial-saddle-trap mitigations landed in v2.6.7d as code patches; the transverse-projection issue was diagnosed in Math290 but the full transverse-projector extension (translations + rotations + internal-phase modes per Math310-AddA) remains pending and is the load-bearing item for F-Pillar6 verdict closure.
6. **Math292 Pillar 6 acceptance:** 4-gauge criterion ($f > 0$, $\Delta F < 0$, $\lambda_{\min}^{\text{transverse}} \geq 0$, `extraction_status = OK`). The 2026-05-01 $N = 16$ striped-seed run delivered $\Delta F < 0$, $f > 0$, but the raw transverse-Lanczos $\lambda_{\min}^{\text{transverse}} \approx -8.51$ does NOT satisfy G3 in the canonical projection scheme; Math310-AddA recorded this as the audit-flag wording correction. The 2026-05-02 $N = 32$ and $N = 64$ Lanczos diagnostics (Math292_transverse_lanczos.py, ARPACK non-convergence after 30k matvecs) further suggest the broken-phase configuration is a saddle manifold of dimension ≥ 4 , requiring a transverse-projector extension (translations + rotations + internal-phase modes) before G3 can be re-evaluated. Math292 valid-acceptance therefore remains *pending*; the 4-gauge criterion is not yet jointly satisfied.
7. **Math236 continuum-limit scan:** Pilot scan at $N = 16$ complete. Full scan at the canonical Pillar-6 ladder $N \in \{16, 32, 64\}$ in progress (Math313 Stage-3 Richardson convention); interim results (2026-05-01) show a convergence pattern consistent with $m_{*,\infty}^2 \approx 0.315$ at the programme-level estimate; the $N = 32$ and $N = 64$ data points required for a closed Richardson certificate remain in iterative refinement [?].

Known limitations:

1. Zero-mode projector is built once from initial seed Ψ_0 , not rebuilt per Newton step. Consequence: at early steps the projected gradient is approximate. Mitigation: trust-region mechanism provides robustness.
2. Class-II sector uses numerical Hessian-vector (central differences), introducing $\sim 1\%$ asymmetry. Mitigation: GMRES handles non-symmetric matrices; CG not recommended.
3. Phase 3 (free-energy comparison) only checks against $\Psi = 0$ vacuum, not against competing branches (lamellar, cylindrical). A full comparison would require solving additional PDEs.

V. DISCUSSION AND OUTLOOK

The solver constitutes the current developmental numerical infrastructure for TECT. The first $\Delta F < 0$ broken-energy data point ($N = 16$, 2026-05-01) demonstrates that the solver reaches the broken-phase basin and delivers consistent per-step diagnostics; full Math292 valid-acceptance, however, remains pending the transverse-projection re-certification and the $N = 32/N = 64$ ladder.

Near-term improvements (Tasks #115–#120):

1. Pytest module for routing-layer contracts (Task #115): three Math74 Addendum-A contracts (R'_1, R'_2, R'_3) to be validated at the integration layer.
2. GPU CUDA backend (Task #54): offload the `real_backend_pt_bcc_mixed_v3` residual and Hessian-vector evaluations to GPU, enabling $N = 64$ runs on commodity hardware.
3. Adaptive mesh refinement (exploratory): permit non-uniform lattice spacing to focus resolution near order-parameter gradients, reducing total DOFs.

The solver is now available for audited developmental use in Pillar-level numerical evidence generation, with full production-grade status deferred until the Math292 transverse-projection re-certification and the $N \in \{16, 32, 64\}$ ladder closure are complete. All subsequent TECT runs cite this auxiliary paper and commit their diagnostics to the Runs directory with full provenance.

ACKNOWLEDGMENTS

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- J. Lee, TECT-Math73, TECT internal note (2026).
- J. Lee, TECT-Math236 striped-seed run, TECT internal note (2026).
- J. Lee, TECT-Math236: continuum-limit-scan numerical implementation, TECT internal note (2026).